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COLLEGE OF ENGINEERING
POONJAR

GROUP 1

PROJECT

SYNAPSE

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FEASIBILITY STUDY

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FEASIBILITY STUDY

This feasibility study evaluates the project against four standard dimensions — technical, economic, operational, and schedule — using the requirements defined in Phase 2 Requirement Analysis and the evidence gathered in the literature survey as its basis, rather than treating feasibility as a separate, unsupported judgement call.

TECHNICAL FEASIBILITY

2. Dataset Availability

Historical running data for individual Indian Railways trains is obtainable from public running-status sources; the RSTGCN study (Chowdhury et al., 2025) confirms this directly, having scraped and released a nationwide operational dataset from `runningstatus.in` covering scheduled/actual arrival and departure times, station codes, and delays. Smaller, ready-made Indian Railways delay datasets are also available on public data platforms such as Kaggle. For a two-to-three-train pilot, this is more than sufficient — the project's data requirement is a small subset of what RSTGCN and Banerjee et al. (2025) each already assembled for a much larger scope.

Weather data is available through free-tier APIs such as OpenWeatherMap, and public-holiday/festival calendars for India are freely published and machine-readable. No contextual data source required by FR-3 depends on paid or restricted access at pilot scale.

2. Software and Technical Skill Availability

The software stack specified in Section 2.4 — Python, Scikit-learn, XGBoost, LightGBM, Pandas, NumPy, and Flask is open source, well documented, and was the same stack (or a close variant of it) used in eight of the ten papers reviewed in Sub-Phase 3, which is itself evidence that the required skill set is learnable from existing tutorials and academic material rather than requiring specialist training. The skills involved — Python programming, supervised machine learning (regression and classification), basic data cleaning/feature engineering, and a simple Flask web interface — fall within a standard B.Tech curriculum and are achievable within the project timeline.

3. Hardware Availability

The hardware specification in Section 2.4.2 (an Intel i5-class processor, 8 GB RAM, 5–10 GB storage) matches a typical personal or lab laptop. Where local hardware is insufficient for a particular training run, Google Colab provides free GPU/CPU compute, which is already listed as the project's experimentation environment. No specialised or high-performance computing infrastructure is required at the pilot scale defined in the project's system constraints (Section 2.6).

Conclusion

All four technical prerequisites — data, software, skills, and hardware — are satisfied using freely available resources and skills already within scope for the project team. The project is judged to be technically feasible.

ECONOMIC FEASIBILITY

Cost Category	Estimate
Development (labour)	None (student project — no salaried development time)
Programming language & ML libraries	\$0 — Python, Scikit-learn, XGBoost, LightGBM, Pandas, NumPy are all open source
Backend framework	\$0 — Flask is open source
Development environment	\$0 — VS Code and Google Colab / Jupyter Notebook are free

Version control	\$0 — Git and GitHub (public/free-tier repository)
Weather API	\$0 at pilot scale — OpenWeatherMap's free tier covers the pilot's call volume; a paid tier would only become relevant at national scale
Hardware	\$0 incremental — existing personal/lab laptop meets the Section 2.4.2 specification
Deployment / hosting	\$0 — Render or PythonAnywhere free tiers are sufficient for a two-to-three-train demo deployment
Maintenance (periodic retraining)	Minimal — compute cost of retraining a tree-based model on a small dataset is negligible on free-tier infrastructure

The only cost that would grow materially is API and hosting cost if the system were scaled from a pilot of two to three trains toward the pan-India coverage identified as future scope at that point, OpenWeatherMap's free-tier call limits and a free hosting tier's resource caps would likely be exceeded, and paid tiers or institutional infrastructure would need to be budgeted for. At the scope defined for this project, however, the economic case is straightforward: the tooling is free, and the main cost is development time already accounted for as project effort.

OPERATIONAL FEASIBILITY

1. Ease of Use

Per NFR-5, the system asks a user for only four pieces of information — train number, journey date, boarding station, and destination station — and returns its prediction in plain language (predicted delay, confidence score, contributing factors). This matches the level of interaction a passenger already has with existing railway apps, so no new user behaviour needs to be learned.

2. Benefits to Passengers and to Railway Authorities

For passengers, the system's core benefit is direct: a station-wise forecast rather than a current-status-only view lets a traveller plan connecting transport and appointments with more confidence, addressing the problem statement in Section 1.3. For railway authorities, the value is indirect but consistent with what the literature survey found — Sharma et al. (2024) and Banerjee et al. (2025) both frame the same kind of model as a tool for identifying operational bottlenecks and delay-prone patterns, which is a natural secondary use of the same prediction pipeline even though this project's primary interface is passenger-facing.

3. Practical Implementation

Restricting the pilot to two or three trains keeps the operational footprint small and testable: a working prototype can be validated end-to-end before any scale-up decision is made. This mirrors the trajectory the literature itself points to — Banerjee et al. (2025) explicitly call for pan-India coverage as the next step beyond their 20-train study, and RSTGCN (Chowdhury et al., 2025) demonstrates that full-network coverage is achievable once a project is ready for it. Starting small and scaling only after validation is therefore not just a scope-management choice for this project, but the same path the two most directly comparable studies in the survey have already taken.

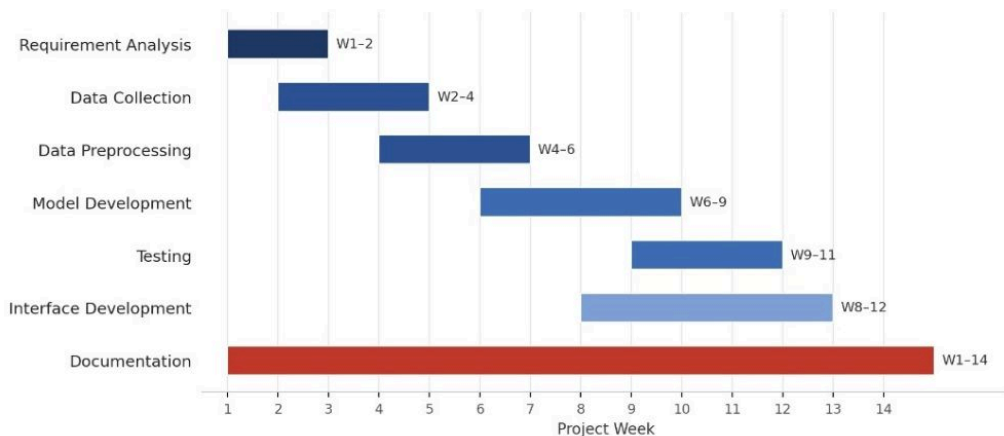
One caveat is worth stating plainly: because the system is advisory only and does not integrate with actual railway dispatch or signalling systems, no coordination with railway operations is required to build and demonstrate it. A future production deployment serving real passengers at scale, however, would need data-sharing arrangements with Indian Railways or a licensed data provider, which is outside this project's current scope but does not block the pilot.

Conclusion

The system is operationally feasible at pilot scale: it requires no new user behaviour, delivers a clear benefit to its primary users, and can be implemented and demonstrated without requiring access to, or agreements with, railway operational systems.

SCHEDULE FEASIBILITY

Figure below sets out an indicative 14-week schedule covering the seven activities identified for this phase. Data collection and preprocessing are sequenced to finish before model development begins in earnest, while interface development is scheduled to overlap with the later stages of model development and testing — the frontend does not depend on a finished model, only on a stable prediction API contract, so building it in parallel shortens the overall timeline rather than lengthening it. Documentation runs throughout, reflecting that the requirement, literature, and feasibility work in this phase is itself part of the project's documentation output, not a task saved for the end.



Activity	Duration	Notes
Requirement Analysis	Weeks 1–2	Functional/non-functional requirements, system requirements (this document set)
Data Collection	Weeks 2–4	Historical train data, weather data, holiday/festival calendars for the pilot trains
Data Preprocessing	Weeks 4–6	Cleaning, missing-value handling, encoding, feature engineering
Model Development	Weeks 6–9	Training and comparing Random Forest, XGBoost, Gradient Boosting, LightGBM
Testing	Weeks 9–11	Model evaluation (MAE, RMSE, R^2), validation against held-out data
Interface Development	Weeks 8–12	Flask backend, web form, results display (overlaps model development/testing)
Documentation	Weeks 1–14	Continuous — requirement, literature, feasibility, and final reporting

This schedule fits comfortably within a standard academic semester. The main schedule risk is data collection (Weeks 2–4): if historical records for the chosen pilot trains turn out to be harder to assemble than the sources identified in Dataset availability suggest, the two-week buffer built into the gap between data preprocessing finishing (Week 6) and model development's later stages (through Week 9) provides some room to absorb a short delay without moving the final documentation deadline.

Overall Feasibility Summary

Across all four dimensions, the project is judged feasible at the pilot scale defined in Requirement Analysis: the required data, tools, and skills are available now and at no material cost; the system offers a clear, low-friction benefit to its primary users without requiring coordination with railway operations; and the seven-activity schedule fits within a normal academic timeline with an identified point of schedule risk and buffer. This feasibility assessment, together with the requirement specification and literature survey, completes the analysis this phase set out to produce and provides the basis for the documentation compilation in Phase 5.

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